

ASHISH SAH

R&D Software Engineer, Ideaforge

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EXPERIENCE

Ideaforge Technology Ltd.

R&D Software Engineer I

04 Nov 2024 - Working Bengaluru, Karnataka

- Working on Flyght Cloud which is all-in-one drone data management platform that empowers industries and government projects with drone-based mapping, 2D-3D and analytics solutions. I am working as a full stack developer and create REST APIs in Rust Axum and React.js in frontend.
- Developed an end-to-end geospatial machine learning pipeline to detect stockpiles from high-resolution DEM data using Python. Engineered terrain-based features such as slope and multi-scale elevation differences, and generated labeled datasets by rasterizing GeoJSON annotations. Trained a Random Forest classifier on large-scale sampled data with memory-efficient chunk processing, and implemented block-wise inference for scalable predictions. Applied morphological post-processing to refine outputs and converted raster predictions into GeoJSON polygons for integration with GIS workflows.
- Created a tool that made FlyghtCloud application SEO friendly using SSG taking inspiration from React Snap, without adding platform like Next.js.
- Lead the frontend testing initiative using JEST, auto-generation of unit test cases using LLM. Set up of the environment to do end to end testing covering up the important flows of the application, reducing the real time major production works and increasing the team collaboration without breaking existing features.
- Worked on Three JS by customizing the Potree.js library for the company use case. Worked on CesiumJs to integrate the 3D models in the browser.
- Implemented Mission Planner web map drone viewer simulator, with local map server provider, used mathematical algorithms for the path planning. React JS Express JS and JSON server..
- Helped in migration from fluid UI to modern AntD setting up HTML/CSS standards using Styled components. Contributed to realtime drone mapping system using Open layers JS.
- Worked on drone management tool which deals with task planning, testing and approvals of drones. With the backend in Node JS (Hapi Js) and frontend in Next, I contributed with creating multiple asynchronous RESTful services and React components. Worked on code optimisation and reusability of components. Revamped different modules for performance optimisation. Active participation in code reviews and technical discussions.
- Technologies: **Next JS, React.js., Node.js, Rust, Python, Figma, Openlayers, Leaflet JS, MongoDB, Postgress, Cesium JS, Potree, Docker, Jest, Test Automation, Photogrammetry, Scikit-learn, Random Forest, Rasterio, NumPy, Pandas**

Tata Steel

Senior Associate

04/04/2018 - 12/10/2024 Jamshedpur

- Contributed to the development and maintenance of industrial-grade web applications, combining expertise in React.js, Node.js, and Express.js to build scalable, maintainable, and production-ready modules.
- Collaborated on an Interlock Bypass Management System and an IoT-based real-time monitoring dashboard, where performance, usability, and reliability were essential. These projects involved real-time data streaming, visualization, and REST APIs, and were deployed with industry-grade practices.
- Co-developed an employee-facing e-learning platform using jQuery, Bootstrap, and Express.js, delivering a smooth UX with intuitive design patterns. Also contributed to front-end design tasks such as translating wireframes and mockups (PSD/Sketch) into clean, semantic HTML/CSS code.
- Worked alongside cross-functional teams and tech leads to design and implement critical sub-modules, while following best

PROFESSIONAL SUMMARY

For more than six years, I have been a contributor in the vast landscape of the web, specializing in full-stack development and automation using JavaScript, Rust, and Python. I enjoy the union of computer science and mathematics.

SKILLS

Full Stack Developer :

JavaScript, RUST, Node.js TypeScript, Next.js, React.js, Express.js, redux, zoom-sdk, django, MongoDB, Postgres, Grafana, In-dexDB, Cesium JS, OpenLayers, Rust Axum, Hapi JS, Next JS, Docker, Fast API, Python, PyMavlink,Antd, Figma, Tailwind CSS, JEST, Three JS

EDUCATION

MCA - Masters of Computer and Application

Srintath university

12/6/2022 - 12/6/2024 Adityapur, Jharkhand, India

PROJECTS

Study Send : Upload your study materials, generate AI-powered exams and revision cards from any document, then earn from paid sessions helping others who are stuck on the same topics.

- Created RAG and vector embedding to generate notes Summarization using Llama, Open AI SDK
- Integrated zoom video SDK to provide support for the live calls among students.
- Automatic revision flash point card, mock exams generation using ML and AI.
- Created Migration of database using drizzle and sql to types npm library.
- Technologies: **Llama, TypeScript, python, aws s3, drizzle ORM, Zoom video SDK, FAST api, RAG, Vectors, AI, pdf.js**
- <https://www.studysend.com/>

Grad AI : Create professional resumes effortlessly with-AI-powered builder.

- 500+ users, 1000+ resumes generated
- Multiple templates for personal portfolio website and resume template
- WYSWYG editor for portfolio and resume
- Multiple free resume templates from known institute on India
- Technologies: **next js, Geminie, Google Gen AI SDK, pdf.js, canvas, prompt chaining, langchain**
- <https://www.gradai.in/>

ACHIEVEMENTS

- Won First prize for web development at Nationals and was second at North Regional Level (HTML,CSS, JS and UI/UX)
- Gold Medal at Tata Ideas (07/2022 - 07/2022) Won this for developing best design for Tata Power Mitra App.
- Gold Medal at Tata Ideas (07/2022 - 07/2022) Tata Sons for developing prototype for dengue and malaria fighting application
- Competent Leader and Communicator at Toastmaster International
- Modern Application Development Course by NPTEL, IIT Madras

practices in coding standards, modular design, and test-driven development. Played a key role in creating and optimizing RESTful services using Node.js, ensuring smooth backend operations and robust integration with frontend components.

- Represented and won First Prize At CII Nationals 2019
- Actively participated at Tata Sons hackathon.
- Technologies: **React, next, javascript, typescript, node js, python**